Math Inquirers

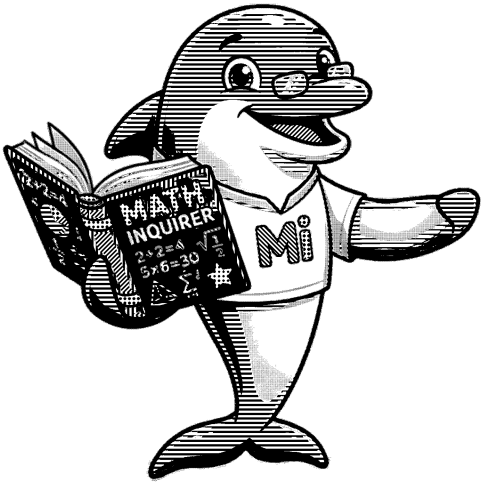
Think. Question. Prove.

Level 1 (ages 6-8) - Term 2

Tutor Answer Key

Book 4

Lessons 16 to 20



FOR THE TUTOR AND GROWN-UPS ONLY - keep away from the child

Contents

Math Inquirers Level 1, Term 2 - Tutor Answer Key, Book 4

Lesson 16 **Skip-Count Tables** **page 3**
Multiplication: Skip-Count Tables

Lesson 17 **page 6**

Lesson 18 **Times Stories** **page 10**
Multiplication And Division: Times Stories

Lesson 19 **Picture Equations** **page 14**
Reasoning About Equality: Picture Equations

Lesson 20 **Find One First** **page 18**
Division And Money: Find One First

Answer key: Lesson 6, Skip-Count Tables

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Count in 5s. Fill in the missing numbers.

Answer: 15, 25, 35

Counting in 5s: 5, 10, 15, 20, 25, 30, 35, 40. The missing numbers are 15, 25 and 35.

Watch for: Writing 14 or 16 (counting in 1s).; Skipping a number.

S2. A frog jumps along the number line. Fill in the blanks.

Answer: 3 jumps; each jump is 5; 3 × 5 = 15 (5 × 3 = 15 is also right)

The frog makes 3 jumps and each jump is 5. 5 + 5 + 5 = 15, so 3 jumps of 5 is 3 × 5 = 15.

Watch for: Counting the numbers on the line instead of the jumps.; Writing 3 + 5 = 8.

S3. Ravi has 4 bundles of 10 straws. How many straws does Ravi have in all?

Answer: 10 + 10 + 10 + 10 = 40; 4 × 10 = 40

4 bundles of 10: 10 + 10 + 10 + 10 = 40. So 4 × 10 = 40 and Ravi has 40 straws.

Watch for: Writing 4 + 10 = 14.; Counting the bundles as 4 straws.

S4. Each bike has 2 wheels. How many wheels do 6 bikes have?

Answer: 12 wheels (6 × 2 = 12; 2 × 6 = 12 is also right)

6 bikes with 2 wheels each: 2 + 2 + 2 + 2 + 2 + 2 = 12. So 6 × 2 = 12.

Watch for: Writing 8 (6 + 2).; Losing count of the bikes.

S5. Fill in the table.

Answer: 10, 20, 30, 40

2 × 5 = 10, 4 × 5 = 20, 6 × 5 = 30 and 8 × 5 = 40.

Watch for: Adding: 2 + 5 = 7.; Writing the number of jumps in the Total box.

S6. Read each statement. Decide if it is True or False.

Answer: True, False, True, False

5 × 4 = 20 is true. 3 × 10 is 30, not 40: false. 6 × 2 = 12 is true. 8 × 5 is 40, not 35: false.

Watch for: Marking 3 × 10 = 40 as true.; Marking 8 × 5 = 35 as true because it ends in 5.

Think harder (page 3)

T1. Look at the dots. The top 2 rows are shaded. Fill in the blanks.

Answer: 2 × 5 = 10; 4 × 5 = 20

2 rows of 5 dots is 2 × 5 = 10. All 4 rows are two lots of 10 dots: 10 + 10 = 20, so 4 × 5 = 20.

Watch for: Writing 10 for both.; Adding 4 + 5 = 9.

T2. Colour the 2s. Ring the 5s. Then fill in the blanks.

Answer: 10, 20, 30, 40

The 2s are 2, 4, 6 ... 40. The 5s are 5, 10, 15 ... 40. The numbers in both are 10, 20, 30 and 40.

Watch for: Listing every number in the 5s.; Missing 40.

T3. Is each number in the 5s table? Circle Yes or No.

Answer: 50 Yes; 27 No; 65 Yes

50 ends in 0: yes. 27 ends in 7: no. 65 ends in 5: yes.

Watch for: Saying 27 is in the table because it is close to 25.; Saying 65 is not in the table because it is a big number.

Go further (page 4)

G1. Here are the 4s up to 40. Ring each number that is also in the 5s table. Then fill in the blanks.

Answer: 20 and 40

The 4s are 4, 8, 12, 16, 20, 24, 28, 32, 36, 40. The ones that end in 5 or 0 are 20 and 40.

Watch for: Ringing 16 or 24 (they do not end in 0 or 5).; Ringing only 20.

G2. Count in 3s up to 30. Fill in the missing numbers. Then find the numbers in both the 3s and the 5s.

Answer: 12, 18, 24; 15 and 30

The 3s are 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. The 5s up to 30 are 5, 10, 15, 20, 25, 30. The numbers in both are 15 and 30.

Watch for: Writing 11 or 13 (counting wrongly after 9).; Writing 10 for a number in both tables.

G3. Ali jumps 5 each time. Mei jumps 4 each time. They both want to reach 40.

Answer: Ali 8 jumps, Mei 10 jumps, 2 more

8 jumps of 5 make 40 (8 × 5 = 40). 10 jumps of 4 make 40 (10 × 4 = 40). 10 − 8 = 2 more jumps.

Watch for: Guessing the number of jumps of 4 without counting in 4s.; Forgetting to subtract at the end.

Investigation (page 5): Last digits

10 and 20 are in both tables (15 and 25 are only in the 5s). Both end in 0. The 5s end in 5 or 0 and the 2s end in 0, 2, 4, 6 or 8: the only digit in both lists is 0.

Look for: A child who tests several numbers, writes the last digits of both tables and sees that only 0 is in both lists. Credit any clear reason. Do not mark a child down for stopping at a few tests.

Exit ticket (page 5)

E1. Count in 5s. Fill in the missing numbers.

Answer: 15 and 25

Counting in 5s: 5, 10, 15, 20, 25, 30.

Watch for: Writing 11 or 12.; Skipping a number.

E2. Which sentence is true? Write its number in the bracket.

Answer: (2) 4 × 10 = 40

4 × 10 = 40 (4 jumps of 10). 3 × 5 is 15, not 20; 6 × 2 is 12, not 14; 8 × 5 is 40, not 35.

Watch for: Choosing 8 × 5 = 35.; Choosing 3 × 5 = 20.

E3. Ben puts 5 stickers on each of 6 pages. How many stickers does Ben have in all?

Answer: 6 × 5 = 30 (5 × 6 = 30 is also right); 30 stickers

6 pages with 5 stickers on each page: 6 jumps of 5 land on 30.

Watch for: Writing 6 + 5 = 11.; Writing 6 × 5 = 11.

Sneak-peek puzzles (not in the workbook)

1. 20
2. 10 and 20. Ali says every second number and Mei says every fifth number, and they both say 10 and 20.

Skip-Count Tables: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 8 and 12
Counting in 2s: 2, 4, 6, 8, 10, 12.

2. 10, 20, 30
Counting in 5s: 5, 10, 15, 20, 25, 30.

3. 30
3 jumps of 10: 10, 20, 30.

4. 10 and 20
The 5s are 5, 10, 15, 20. The 2s include 10 and 20 but not 5 or 15.

5. 28
4 × 7 is two lots of 2 × 7: 14 + 14 = 28.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 7, Sharing and Grouping

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Share 8 dots equally among the 4 plates. Draw the dots. Then fill in the blank.

Answer: Each plate has 2 dots

Deal one dot at a time to each of the 4 plates. After 2 rounds all 8 dots are drawn. Each plate has 2 dots, and 4 groups of 2 is 8.

Watch for: Putting all the dots on one plate.; Writing 4, the number of plates.

S2. Mei packs 12 cookies into bags. She puts 3 cookies in each bag. Draw a ring around each group of 3. Then fill in the blank.

Answer: Mei needs 4 bags

12 cookies in groups of 3: ring 3, ring 3, ring 3, ring 3. That is 4 groups, so Mei needs 4 bags. Count in jumps of 3 to check: 3, 6, 9, 12.

Watch for: Writing 3, the number in each bag.; Ringing groups of different sizes.

S3. Ali has 14 sweets. He shares them equally among 2 friends. How many sweets does each friend get?

Answer: Each friend gets 7 sweets

Dealing 14 sweets to 2 friends one at a time gives 7 each. Check: 2 groups of 7 is 14.

Watch for: Writing 2, the number of friends.; Writing 14 or 16 by adding.

S4. Share the sweets equally among the friends. Fill in the table.

Answer: 4, 3, 4

8 sweets among 2 friends is 4 each. 12 sweets among 4 friends is 3 each. 20 sweets among 5 friends is 4 each. Check: 2 groups of 4 is 8, 4 groups of 3 is 12, 5 groups of 4 is 20.

Watch for: Writing the number of friends in the last column.; Giving 5 for the last row because it has a 5.

S5. Ben puts 20 pencils into pots. He puts 5 pencils in each pot. How many pots does Ben use?

Answer: Ben uses 4 pots

5, 10, 15, 20 is 4 jumps, so Ben uses 4 pots. Check: 4 groups of 5 is 20.

Watch for: Writing 5, the number in each pot.; Stopping at 15.

S6. Mei shares 12 stickers equally among 4 friends. Each friend gets 3 stickers.

Answer: 4 × 3 = 12 (3 × 4 = 12 is also right); Yes

4 friends with 3 stickers each is 4 groups of 3: 4 × 3 = 12. (3 × 4 = 12 is also right.) That is the 12 stickers Mei began with, so the answer matches.

Watch for: Writing 4 × 3 = 7 by adding.; Circling No because the numbers look different.

S7. Read each story. Circle Share or Group.

Answer: Share; Group

Lily deals the beads out equally to 3 friends, so it is Share (4 beads each). Jamie puts 6 beads in each box, so it is Group (2 boxes).

Watch for: Choosing Share because both stories have 12 beads.; Choosing by the number 3 or 6.

Think harder (page 3)

T1. Ali has 19 counters. He puts them into groups of 4.

Answer: 4 groups; 3 left over

4, 8, 12, 16 is 4 groups, and 16 counters are used. 19 - 16 = 3 counters are left. Three is not enough for another group of 4.

Watch for: Writing 5 groups by rounding up.; Forgetting the counters left over.

T2. Ben shares 20 biscuits equally with his 3 cousins. Aisha says 3 people share the biscuits.

Answer: No; 5 biscuits each

Ben shares with his 3 cousins, so Ben is one of the sharers: Ben and 3 cousins is 4 people, not 3. 20 biscuits among 4 people is 5 each, because 4 groups of 5 is 20. (A story that says 'shares among 3 friends' does not count the sharer: there the 3 friends are all the people.)

Watch for: Saying Yes because the story says 3 cousins.; Sharing 20 among 3 people.

T3. Mei has 18 sweets. Read the two stories.

Answer: Group; 3 bags; Share; 3 sweets

Story A tells us 6 sweets in each bag, so it is Group, and 18 in groups of 6 makes 3 bags. Story B tells us 6 bags, so it is Share, and 18 shared among 6 bags is 3 in each bag. Both answers are 3, but the meanings are different.

Watch for: Saying Share for both stories.; Answering 6 for (b) or (d).

T4. Ravi has 17 stickers. He puts 5 stickers in each bag.

Answer: 3 bags; 2 left over; 3 more

3 bags use 15 stickers, so 2 stickers are left over. One more bag needs 5 stickers, and 2 + 3 = 5, so he needs 3 more.

Watch for: Writing 2 for (c), the number left over.; Writing 4 bags.

Go further (page 4)

G1. Jenny has 12 sweets. She shares them equally among some friends. Fill in the table. Then answer (a) and (b).

Answer: 12, 6, 4, 3, 2, 1; No; 2 left over; 18 sweets: 1, 2, 3, 6, 9, 18 friends

1 friend gets all 12. 2 friends get 6 each. 3 friends get 4 each. 4 friends get 3 each. 6 friends get 2 each. 12 friends get 1 each. With 5 friends, each gets 2 and that uses 10 sweets, so 2 are left over and the answer to (a) is No. (c) With 18 sweets: 1 friend gets 18, 2 get 9, 3 get 6, 6 get 3, 9 get 2 and 18 get 1. Try 4, 5, 7 and 8: each leaves some over. So the numbers are 1, 2, 3, 6, 9 and 18, six ways in all. Finding every way to share is the start of factors, which school meets later.

Watch for: Writing the number of friends instead of the number each.; Saying Yes for 5 friends.; Part (c): stopping at 2, 3 and 6 and missing 9 or 18, or writing how many each gets instead of the number of friends.

G2. Jenny shares 20 pencils equally among 4 pots. Then she puts 3 more pencils in each pot.

Answer: 5 pencils; 8 pencils; 32 pencils

20 pencils among 4 pots is 5 in each pot. Then 3 more in each pot makes 5 + 3 = 8 in each pot. 4 pots with 8 pencils is 8 + 8 + 8 + 8 = 32 pencils in all.

Watch for: Adding 3 to 20.; Stopping after 5.

G3. Wei has 20 stickers. He keeps 2 stickers. He shares the rest equally among 3 friends.

Answer: 18 stickers; 6 stickers each

20 - 2 = 18 stickers are shared. 18 shared among 3 friends is 6 each, because 3 groups of 6 is 18.

Watch for: Sharing all 20 stickers.; Writing 7 by sharing 20 among 3.

Investigation (page 5): Sharing among 4 plates

Left over: 1, 2, 3, 0, 1, 2, 3, 0. The left-over numbers 0, 1, 2, 3 repeat. Never more than 3 are left over, because with 4 left over we could give one more counter to every plate.

Look for: A child who deals out the counters for several numbers, writes the left-over numbers and notices that they repeat 1, 2, 3, 0. Credit a child who says the same numbers come back every 4 counters, or who says the left over is never as big as the number of plates. Do not mark down a child who finishes only some of the numbers.

Exit ticket (page 5)

E1. 20 sweets are shared equally among 4 children. Fill in the blank.

Answer: 5

Deal 20 sweets to 4 children one at a time: 5 each. Check: 4 groups of 5 is 20.

Watch for: Writing 4, the number of children.; Writing 24 by adding.

E2. Which story puts 12 sweets into groups of 4? Write its number in the bracket.

Answer: (2) Mei puts 12 sweets into bags of 4.

In story 2 the sweets are put into bags of 4, so they are in groups of 4 and there are 3 bags. Story 1 shares among 4 friends (3 each). Stories 3 and 4 do not make groups.

Watch for: Choosing story 1 because it has a 4 in it.; Choosing story 3 because it takes away 4.

E3. Ali puts 18 cards into piles. Each pile has 3 cards.

Answer: 6 piles; 6 × 3 = 18

Count in jumps of 3 to 18: 3, 6, 9, 12, 15, 18 is 6 jumps, so 6 piles. Check: 6 × 3 = 18.

Watch for: Writing 3, the number in each pile.; Writing 15 by stopping early.

Sneak-peek puzzles (not in the workbook)

1. 3
2. Mei. Ali puts 3 sweets in each bag (4 bags). Mei shares 12 among 3 bags, so each bag has 4.

Sharing and Grouping: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 5
10 stickers dealt to 2 friends one at a time is 5 each. Check: 2 groups of 5 is 10.

2. 4 and 4
12 pencils in groups of 3 is 4 pots (3, 6, 9, 12). Check: 4 groups of 3 is 12.

3. Group
The story says how many apples are in each box (5), so Amelia puts the apples into groups of 5. There are 3 boxes.

4. 3 groups, 1 left over
4, 8, 12 is 3 groups. 13 - 12 = 1 counter is left over.

5. 3
18 - 6 = 12 stickers are left. 12 shared among 4 friends is 3 each, because 4 groups of 3 is 12.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 8, Times Stories

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Mei has 3 bags. Each bag has 5 sweets. Write the number in each box of the strip. Then fill in the blank.

Answer: 5, 5, 5 in the boxes; 15 sweets

There are 3 bags, so 3 boxes, and each box is 5 sweets. 5 + 5 + 5 = 15, so Mei has 15 sweets in all.

Watch for: Writing 3 in the boxes.; Adding 3 + 5 = 8.

S2. Ben has 4 boxes. Each box has 3 pencils. How many pencils does Ben have in all?

Answer: 4 groups of 3 = 12 (3 groups of 4 = 12 is also right); 12 pencils

4 boxes with 3 pencils in each box: 3 + 3 + 3 + 3 = 12. So 4 groups of 3 = 12 and Ben has 12 pencils. Either order gives the right total: if a child writes 3 groups of 4 = 12, ask 'how many boxes?'.

Watch for: Writing 4 + 3 = 7 by adding.; Drawing 3 boxes with 4 in each box (the total is right, so ask 'how many boxes?').

S3. Siti has 5 bags. Each bag has 2 sweets. Which strip shows Siti's sweets? Write its number in the bracket.

Answer: Strip 2

Strip 2 has 5 boxes with 2 in each box: 5 bags of 2 sweets. Strip 1 has 2 boxes of 5. Strip 3 has 5 boxes but the last box has 1, so the boxes are not equal.

Watch for: Choosing strip 1 because it has the same numbers.; Not noticing that the last box of strip 3 is smaller.

S4. Ali has 3 boxes with 5 pens in each box. He also has 2 loose pens. How many pens does Ali have in all?

Answer: 15 pens in the boxes; 17 pens in all

Step 1: 3 boxes of 5 pens is 5 + 5 + 5 = 15. Step 2: 15 and 2 loose pens make 17 pens in all.

Watch for: Writing 5 + 3 + 2 = 10.; Forgetting the loose pens.

S5. Mei has 2 bags. Each bag has 6 sweets. Draw a strip. Then fill in the blank.

Answer: A strip of 2 boxes with 6 in each box; 12 sweets

2 bags with 6 sweets in each bag: draw 2 equal boxes and write 6 in each. 6 + 6 = 12, so Mei has 12 sweets in all.

Watch for: Drawing 6 boxes with 2 in each.; Drawing boxes that are not the same size.

S6. Ravi has 4 bags with 3 marbles in each bag. He loses 2 marbles. How many marbles does Ravi have now?

Answer: 12 marbles in the bags; 10 marbles now

Step 1: 4 bags of 3 marbles is 3 + 3 + 3 + 3 = 12. Step 2: he loses 2, so 12 - 2 = 10 marbles are left.

Watch for: Adding the 2 marbles.; Taking 2 away from 4 or from 3.

Think harder (page 3)

T1. Jenny has 3 boxes with 6 pens in each box and 5 loose pens. She gives 8 pens away.

Answer: 18 pens; 23 pens; 15 pens left

Step 1: 3 boxes of 6 is 6 + 6 + 6 = 18. Step 2: 18 and 5 loose pens make 23. Step 3: 23 take away 8 is 15 pens left.

Watch for: Stopping at 23.; Taking away 8 from 18 before adding the loose pens.

T2. Mei has 7 marbles. Ali has twice as many marbles as Mei.

Answer: 14 marbles; 21 marbles together

Mei's strip is 7. Ali has twice as many: 7 and 7, which is 14. Together they have 7 + 14 = 21 marbles.

Watch for: Writing 9 (7 + 2).; Writing 14 for together.

T3. Mei has 4 stickers. Ravi has twice as many stickers as Mei. Jenny says Ravi has 6 stickers.

Answer: No; 8 stickers

Mei has 4, so Ravi has two strips of 4: 4 + 4 = 8 stickers. Jenny added 2 to 4 to get 6, but twice as many is not 2 more. So Jenny is not correct.

Watch for: Saying Yes because 4 and 2 make 6.; Writing 4 or 12.

T4. Ali has twice as many cards as Mei. Ali has 24 cards. Ali gives some cards to Mei. Then they have the same number of cards.

Answer: Mei 12 cards; Ali gives 6 cards

Ali has two strips of 12, so Mei has 12. The gap is 12. Ali gives half of the gap: two equal parts of 12 are 6 and 6, so he gives 6. Then Ali has 24 - 6 = 18 and Mei has 12 + 6 = 18. They have the same number.

Watch for: Giving all 12 cards.; Writing 12 for (b).

Go further (page 4)

G1. Wei puts 3 stickers on each of 4 cards. He puts 2 stickers on each of 5 other cards. He also has 4 stickers in his hand.

Answer: 12 stickers; 10 stickers; 26 stickers at first

4 cards with 3 stickers: 3 + 3 + 3 + 3 = 12. 5 cards with 2 stickers: 2 + 2 + 2 + 2 + 2 = 10. 12 + 10 + 4 = 26 stickers at first (the 4 in his hand count too).

Watch for: Forgetting the 4 stickers in his hand.; Using 3 for both sets of cards.

G2. Siti has 5 marbles. Ravi has 3 times as many marbles as Siti.

Answer: 15 marbles; 20 marbles altogether

Siti's strip is 5. Ravi has 3 times as many: 5 + 5 + 5 = 15. Together: 5 + 15 = 20 marbles.

Watch for: Writing 8 (5 + 3).; Writing 15 for altogether.

G3. Kumar has 3 times as many pens as Lily. They have 24 pens together.

Answer: Lily 6 pens; Kumar 18 pens

Lily has 1 box and Kumar has 3 boxes, so there are 4 equal boxes. 24 pens in 4 equal boxes is 6 in each box (6 + 6 + 6 + 6 = 24). Lily has 6 and Kumar has 18. Check: 3 × 6 = 18 and 6 + 18 = 24.

Watch for: Writing 8 (24 shared into 3).; Writing 3 times 24.

Investigation (page 5): Twice, and twice again

Mei 3, Ali 6, Ben 12, Jenny 24. Each child has twice as many as the child before, so each strip is twice as long as the one before. The numbers double each time: 3, 6, 12, 24.

Look for: A child who draws or builds each strip as two copies of the strip before, writes 3, 6, 12, 24 and notices that each number is double the last. Credit a child who says the strips get twice as long, or who checks 12 + 12. Do not mark down a child who finds only Ali and Ben.

Exit ticket (page 5)

E1. Ali has 4 bags. Each bag has 5 sweets. Which number sentence finds how many sweets Ali has in all? Write its number in the bracket.

Answer: (2) 4 × 5

4 bags with 5 sweets in each bag is 4 groups of 5, which is 4 × 5. Adding 4 + 5 joins the numbers of bags and sweets. 5 - 4 finds a gap.

Watch for: Choosing 4 + 5.; Choosing 5 - 4.

E2. Lily has 2 bags with 5 marbles in each bag. She also has 3 loose marbles. How many marbles does Lily have in all?

Answer: 13

Step 1: 2 bags of 5 marbles is 10. Step 2: 10 and 3 loose marbles make 13 marbles in all.

Watch for: Writing 10 (forgetting the loose marbles).; Writing 2 + 5 + 3 = 10.

E3. Ravi has 3 bags with 4 sweets in each bag. He eats 2 sweets. How many sweets does Ravi have now?

Answer: 10

Step 1: 3 bags of 4 sweets is 12. Step 2: he eats 2, so 12 - 2 = 10 sweets are left.

Watch for: Adding the 2 sweets: 14.; Writing 12 (forgetting the 2 he eats).

Sneak-peek puzzles (not in the workbook)

1. 12
2. 15. Ben has twice as many as Mei: 5 and 5, which is 10. Mei has 5 and Ben has 10, so together they have 15.

Times Stories: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 4 × 2 = 8 (or 2 × 4 = 8, either order) and 8 sweets
4 bags with 2 sweets in each bag: 2 + 2 + 2 + 2 = 8.

2. 17
3 boxes of 4 pens is 12. 12 and 5 loose pens make 17.

3. 13
4 bags of 4 stickers is 16. 16 take away 3 is 13.

4. 27
Ravi has 9 + 9 = 18 marbles. Together: 9 + 18 = 27.

5. 17
Ali has 4 + 4 + 4 = 12 stickers. Ben has 12 + 5 = 17 stickers.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 9, Picture Equations

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.

Answer: 8

11 - 3 = 8, so the cup holds 8 counters. Check: 8 + 3 = 11.

Watch for: Writing 14 (11 + 3).; Writing 11 (the total).

S2. Each shape is a number. Find the number for the circle.

Answer: 6

The total is 10 and the square is 4. 10 - 4 = 6, so the circle is 6. Check: 6 + 4 = 10.

Watch for: Writing 14 (adding 10 and 4).; Writing 10 (the total).

S3. Each shape is a number. Find the number for the heart.

Answer: 8

8 + 8 = 16, so each heart is 8.

Watch for: Writing 16 (the total).; Writing 32 (adding the total twice).

S4. Each shape is a number. Find the number for the star.

Answer: 5

5 + 5 + 5 = 15, so each star is 5.

Watch for: Writing 3 (counting the stars).; Writing 12 (15 - 3).

S5. Each shape is a number. The square is 4. Fill in the blanks.

Answer: 10; 5

14 - 4 = 10, so the 2 triangles make 10. The triangles are the same, so each is 5. Check: 5 + 5 + 4 = 14.

Watch for: Writing 14 for each triangle.; Writing 10 for each triangle.

S6. Ben says the circle is 7. Look at the picture.

Answer: No; 6

7 + 7 = 14, but the line says 12. So Ben is not correct. 6 + 6 = 12, so the circle is 6.

Watch for: Saying Yes because 7 is close to 6.; Writing 12 for the circle.

S7. Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.

Answer: 10; 5

13 - 3 = 10 sweets are in the 2 bags. 5 + 5 = 10, so each bag has 5 sweets. Check: 5 + 5 + 3 = 13.

Watch for: Writing 13 for each bag.; Writing 6 (13 shared between 2 bags, ignoring the loose sweets).

Think harder (page 3)

T1. Each shape is a number. Find the number for each shape. Show your working.

Answer: square 6; heart 7

Line 1 has one more square than line 2, and 19 is 6 more than 13, so the square is 6. Then 6 + heart = 13, so the heart is 7. Check: 6 + 6 + 7 = 19.

Watch for: Writing 19 and 13 as the answers.; Finding the square but forgetting the heart.

T2. Kumar says the heart is 8 and the star is 7. Look at the picture. Decide if each statement is True or False.

Answer: True; False

Line 1: 8 + 7 = 15, which matches. Line 2: 8 + 8 + 7 = 23, but the line says 21. Kumar's numbers work in line 1 only.

Watch for: Checking only line 1.; Saying line 2 is true because 23 is close to 21.

T3. Each shape is a number. Find the number for each shape.

Answer: heart 3; circle 5

The 2 hearts make 6, so the heart is 3. Then 3 × circle = 15 means 3 groups of 5 make 15, so the circle is 5. Check: 3 × 5 = 15.

Watch for: Writing 6 for the heart.; Writing 15 for the circle.

Go further (page 4)

G1. Each shape is a number. Find the number for each shape.

Answer: circle 5; heart 7; star 11

Circle × circle = 25, so the circle is 5 (5 groups of 5). Then heart + 5 = 12, so the heart is 7. Then star - 7 = 4, so the star is 11. Check: 11 - 7 = 4.

Watch for: Writing 12 for the heart.; Taking 4 away from 7 and writing 3 for the star.

G2. Each shape is a number. Fill in the blanks.

Answer: 14; star 8; heart 6

Both lines together have 3 stars and 3 hearts, and 22 + 20 = 42. 3 stars and 3 hearts make 42, so one star and one heart make 14. Then 22 - 14 = 8 is a star, and 14 - 8 = 6 is a heart.

Watch for: Writing 22 and 20 as the answers.; Writing 7 for both shapes.

G3. Each shape is a number. The heart is greater than the triangle. Find the number for each shape.

Answer: heart 5; triangle 4

The pairs that add to 9 are 1 and 8, 2 and 7, 3 and 6, 4 and 5. Only 4 and 5 multiply to 20. The heart is greater, so the heart is 5 and the triangle is 4.

Watch for: Writing 4 for the heart.; Using 3 and 6 (they add to 9 but multiply to 18).

Investigation (page 5): One line or two?

Only circle 4 and triangle 6 fit both lines. One line alone has many pairs that fit (1 and 9, 2 and 8, 3 and 7, 4 and 6, 5 and 5 and more; accept 5 and 5 for line 1). 5 and 5 fails line 2 (5 + 5 + 5 = 15, not 14). The second line is the same as the first with one more circle, and 14 is 4 more than 10, so the circle is 4.

Look for: A child who tries pairs for both lines, finds that only one pair works, or says one line is not enough to find two numbers. Credit any clear reason. Do not mark a child down for stopping at one pair.

Exit ticket (page 5)

E1. Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.

Answer: 9

9 + 9 = 18, so each triangle is 9.

Watch for: Writing 18 (the total).; Writing 2 (counting the triangles).

E2. The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.

Answer: (4) 8

13 - 5 = 8, so the heart is 8. Check: 8 + 5 = 13.

Watch for: Choosing (1) 5, the number of the square.; Choosing (2) 6 or (3) 7 after a counting slip.

E3. Each shape is a number. The star is 5. Find the number for each circle.

Answer: 7

19 - 5 = 14, so the 2 circles make 14. 7 + 7 = 14, so each circle is 7. Check: 7 + 7 + 5 = 19.

Watch for: Writing 14 for each circle.; Writing 19 for each circle.

Sneak-peek puzzles (not in the workbook)

1. 5
2. 3. The two stars make 14, so one star is 7. The heart and 7 make 10, so the heart is 3.

Picture Equations: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 4
4 + 4 = 8, so the star is 4.

2. 9
14 - 5 = 9, so the cup holds 9 counters.

3. 12
4 + 4 + 4 = 12.

4. 3
Line 1 has one more circle than line 2, and 15 - 9 = 6, so the circle is 6. Then 6 + triangle = 9, so the triangle is 3.

5. 5
Line 1: 4 × 4 = 16 (4 groups of 4), so the circle is 4. Line 2: heart + 4 = 9, so the heart is 5.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 10, Find One First

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. 4 pens cost $16. How much does 1 pen cost? Use tokens or draw.

Answer: $4

$16 shared equally among 4 pens is $4 each. Check: 4 groups of $4 make $16.

Watch for: Writing 20 (adding 16 and 4).; Writing 64 (multiplying).; Writing 16 (the total).

S2. 1 pen costs $2. How much do 7 pens cost? Fill in the blanks.

Answer: 14; $14

7 groups of $2 is 2 + 2 + 2 + 2 + 2 + 2 + 2 = 14, so 7 pens cost $14.

Watch for: Writing 9 (adding 2 and 7).; Writing 2 (the cost of 1 pen).

S3. 6 buns cost $12. How much does 1 bun cost? Look at the strip.

Answer: $2

$12 shared equally among 6 buns is $2 each. Check: 6 groups of $2 make $12.

Watch for: Writing 6 (the number of buns).; Writing 72 (multiplying).

S4. 3 books cost $15. Fill in the blanks. Look at the strip.

Answer: $5; $10

$15 shared equally among 3 books is $5 each. 2 books cost 2 groups of $5, which is $10. Check: 3 groups of $5 make $15.

Watch for: Writing 15 for 2 books.; Writing 7 (adding 5 and 2).; Stopping after the first part.

S5. 5 pens cost $15. Lily says 1 pen costs $4. Look at the picture.

Answer: No; $3

5 groups of $4 make $20, not $15. So Lily is not correct. $15 shared equally among 5 pens is $3 each. Check: 5 groups of $3 make $15.

Watch for: Saying Yes because 4 is close to 3.; Writing 15 for 1 pen.

S6. Each book costs $4. Kumar buys 3 books. How much do the 3 books cost? Use tokens or draw.

Answer: $12

3 groups of $4: 4 + 4 + 4 = 12, so the 3 books cost $12.

Watch for: Writing 7 (adding 4 and 3).; Writing 4.

Think harder (page 3)

T1. 4 cups cost $12. Fill in the blanks. Look at the strip.

Answer: $3; $21

$12 shared equally among 4 cups is $3 each. 7 cups cost 7 groups of $3, which is $21. Check: 4 groups of $3 make $12.

Watch for: Writing 84 (12 × 7).; Writing 19 (12 + 7).; Stopping after the first part.

T2. A dozen buns cost $24. Fill in the blanks.

Answer: $2; $6

A dozen is 12, so 12 buns cost $24. $24 shared equally among 12 buns is $2 each. 3 buns cost 3 groups of $2, which is $6.

Watch for: Using 24 shared among 2 buns.; Writing 24 for 3 buns.

T3. Wei pays $24 for 6 such cups. Fill in the blanks.

Answer: $4; $16

$24 shared equally among 6 cups is $4 each. 4 such cups cost 4 groups of $4, which is $16.

Watch for: Writing 24 for 4 cups.; Writing 8 (4 + 4).

T4. 4 pens cost $8. Ben has $10. Fill in the blanks.

Answer: $2; 5 pens

$8 shared equally among 4 pens is $2 each. $10 holds 5 groups of $2, so Ben can buy 5 pens.

Watch for: Writing 4 pens (the number in the story).; Writing 20 (10 × 2).

Go further (page 4)

G1. 2 hats cost $10. 3 scarves cost $18. Jenny buys 1 hat and 2 scarves. Fill in the blanks.

Answer: $5; $6; $17

$10 shared equally among 2 hats is $5 each. $18 shared equally among 3 scarves is $6 each. 2 scarves cost $12. 1 hat and 2 scarves cost $5 + $12 = $17.

Watch for: Adding 10 and 18.; Counting 1 scarf instead of 2.

G2. 8 pens cost $24. Mei has $30. She buys 5 pens. Fill in the blanks.

Answer: $3; $15; $15

$24 shared equally among 8 pens is $3 each. 5 pens cost $15. $30 - $15 = $15 is left.

Watch for: Taking 24 away from 30.; Forgetting to take the cost away from $30.

G3. Shop A sells 4 pens for $12. Shop B sells 5 pens for $20. Which shop sells 1 pen for less?

Answer: $3; $4; Shop A

Shop A: $12 shared among 4 pens is $3. Shop B: $20 shared among 5 pens is $4. $3 is less than $4, so Shop A sells 1 pen for less.

Watch for: Choosing Shop B because 20 is more.; Comparing $12 and $20 without finding 1 pen.

Investigation (page 5): Which pack costs less?

No. In both packs 1 pen costs $4 ($8 shared between 2 pens, and $20 shared among 5 pens), so 1 pen does not cost less in the bigger pack. Finding the price of one lets us compare packs of different sizes.

Look for: A child who finds $3 for both packs, or who says 1 pen costs the same in both. Credit any clear reason. A child who says the bigger pack costs more is right about the total but has not compared one pen.

Exit ticket (page 5)

E1. 3 pencils cost $9. How much does 1 pencil cost?

Answer: $3

$9 shared equally among 3 pencils is $3 each.

Watch for: Writing 12 (adding 9 and 3).; Writing 27 (multiplying).

E2. Each sticker costs $5. How much do 3 stickers cost? Write its number in the bracket.

Answer: (2) $15

Each sticker costs $5, so 3 stickers cost 3 groups of $5, which is $15.

Watch for: Choosing (1) $8 by adding 5 and 3.; Choosing (3) $2 by taking 3 from 5.

E3. 5 cakes cost $10. Fill in the blanks.

Answer: $2; $8

$10 shared equally among 5 cakes is $2 each. 4 cakes cost 4 groups of $2, which is $8.

Watch for: Writing 10 for 4 cakes.; Writing 9 (5 + 4).

Sneak-peek puzzles (not in the workbook)

1. $2
2. $10. 3 apples cost $6, so 1 apple costs $2. 5 apples cost 5 groups of $2, which is $10.

Find One First: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. $2
$10 shared equally among 5 pens is $2 each.

2. 18; $18
6 groups of $3: 3 + 3 + 3 + 3 + 3 + 3 = 18.

3. $6
$12 shared equally among 4 rubbers is $3 each. 2 rubbers cost 2 groups of $3, which is $6.

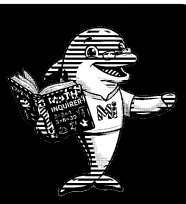
4. $12
A dozen is 12 eggs. $36 shared equally among 12 eggs is $3 each. 4 eggs cost 4 groups of $3, which is $12.

5. $9
1 mug costs $12 shared among 3 mugs, which is $4. 1 plate costs $10 shared between 2 plates, which is $5. 1 mug and 1 plate cost $4 + $5 = $9.

More practice online (no timers, no scores): ask your tutor for the practice link.

TUTOR NOTES • LEVEL 1 • TERM 2 • MI'S CHALLENGE QUEST

Mi's Challenge Quest



8 gates, one from each lesson of the term, all at Start here level, plus one optional secret cave item. 20 minutes after the break, played in pairs or as a group quest with Mi, working on private paper. It is a game: no score or mark is shown to the child, no ranking, no visible timer. You keep private notes only. For the tutor: do not hand these notes to families.

Where it sits. Lesson 20 (the last lesson of the term), minutes 35 to 55, the part Mi's Challenge Quest, straight after the brain break; or at the make-up slot. Owner decisions of 22 Oct 2026: the end-of-term slot is a game with Mi, not a test. The group plays it in pairs (a child and a quest partner) or as a group quest with Mi, on private paper. No score or mark is shown to the child, there is no ranking and no visible timer. The tutor keeps private notes only, for one line to each grown-up and for advice for next term's group. It keeps the 20-minute slot after the break and the question quality: 8 short gates and one optional secret cave item, all at Start here level. Lessons 13 and 14 share gate 3 and Lessons 15 and 16 share gate 4, so every lesson of the term is still visited. Gate 7 asks the child to try a friend's number by putting it back, which is the claim-trying the term line talks about. A middle child needs about 17 minutes of reading aloud and quiet work; a slower child may use a few minutes of the workbook time to finish. The quest sits straight after the brain break (not at the end of the lesson as in Term 1) so children are fresh. The Quest Log has no pictures except the ten-frames in gate 3, because the page generator draws only a few figure types.

1. The gates at a glance

Gate	From	Area (tutor label)	What it uses (Start here level)	Answer
1	L11	Whole Numbers; Addition and Subtraction	A place in a line (1st, 2nd, 3rd) and how many are in front	**5th; 3**
2	L12	Whole Numbers; Addition and Subtraction	A strip for more: the other person and the total	**35; 61**
3	L13	Whole Numbers; Addition and Subtraction	A double from ten-frames (Lesson 13) and odd or even (Lesson 14)	**18; odd**
4	L15	Multiplication and Division; Money	Groups of the same size and the times sign (Lesson 15), counting in 2s (Lesson 16)	**3 × 5 = 15; 8**
5	L17	Multiplication and Division; Money	Share equally and put into groups (Lesson 17)	**4; 5**
6	L18	Multiplication and Division; Money	A times story with a second step (Lesson 18)	**15; 11**
7	L19	Whole Numbers; Addition and Subtraction	A hidden number in a picture equation, and checking a friend's number (Lesson 19)	**No; 9**
8	L20	Multiplication and Division; Money	Find one first, then many (Lesson 20)	**$2; $10**
Secret cave	L11	Whole Numbers; Addition and Subtraction	Queue counting: 8th from the front and from the back, and checking a claim (optional)	**No; 15 children (7 in front of Ravi, Ravi, and 7 behind him)**

The area labels are for the tutor only and never appear on a child or parent page. Each gate is a fresh example of the Start here idea of its lesson (numbers chosen so that they do not repeat a lesson page), so a child who followed the lesson can open it. Before first use, read each gate against what the lesson actually taught.

2. Playing it: before and during

1. A week before: send the grown-up note (L20_Quest_Parent_Note.pdf) and the Quest Log (L20_Quest_Log.pdf, 3 pages) with the lesson materials. Families print the Quest Log at 100%, keep it closed in an envelope and find a quest partner (a grown-up or a brother or sister). A family with no printer tells you: arrange another way (for example the make-up slot).
2. Tell the group in the opening and at slide s-quest1: this is Mi's Challenge Quest, a game with Mi. Everyone works on their own paper with a quest partner beside them. It is not a race, a gate may be skipped, and nothing is shown to the group.
3. Start at minute 35 (straight after the brain break) and finish by minute 55. Children open the Quest Log only when you say. The paper lies flat on the table and is never held up. The quest partner may talk about how to start and ask 'how do you know?', but

does not give answers. Cameras may be on or off. Use your own watch out of sight of the camera: no clock, countdown or timer is ever shown to the children.

4. Read each page aloud, one gate at a time (the script has the words), then leave quiet working time: about 3 minutes for page 1, 3.5 for page 2 and 4 for page 3. After each page say that the gates are open and Mi swims on, whatever the answers. Say nothing about right or wrong, and read no answers. A middle child needs about 17 minutes in all; a slower child may finish in the first minutes of the workbook.
5. Allowed: fingers, counters and talk with the quest partner. Not allowed: a calculator, a grown-up giving answers, looking at the workbook. A child who finishes early turns the Quest Log over and draws quietly.
6. After page 2, take a one-minute stretch (stand, shake out, sip water). After page 3, show slide s-quest2. Do not collect or look at any Quest Log. While children work, notice (do not write on the shared screen) how they start: counters, drawing, strip, talking to the partner.

3. After the lesson: photos and your private notes

1. Same day: the grown-up takes ONE clear photo of each of the 3 pages (page flat, taken straight from above, the whole page in view, and the name on each page) and sends them to the program's business WhatsApp, +65 9880 5962, and to no other number. Never to a child, never to a personal phone, never into the class chat.
2. Within 3 days: open the photos only on the business WhatsApp (never on a personal phone) and write your private notes (section 5) for each child, from the photos and what you saw on camera. Section 4 shows what each answer tells you; accept 5 or 5th for gate 1 part a, and either order of 3 and 5 in gate 4. Nothing is written on the child's paper, nothing is added up and no order of children is made. Nothing about the quest goes back to the child or the group except thanks.
3. Write ONE line for each grown-up from your notes (section 7): what we saw, in plain words, and one small next step if it helps. It is sent to the parent only. No marks, no x out of 8, no ranking, no comparison with other children, no prediction.
4. Fill in the group advice sheet (section 6): what several children were still working on, with no names, and what you will do in the first lessons of next term. It is kept for next term's tutor and is never shown to families.
5. Once the line is sent, save the photos to the program's private file store and delete them from the business WhatsApp chat. The program keeps the photos for 90 days after the program ends, then deletes them, as the privacy notice says. Delete each child's notes sheet when the line is sent (keep only the line in the family file, and the group advice sheet). Never forward the photos or save them to a personal phone or any other device.
6. At the start of next term say to the whole group: thank you for playing Mi's Challenge Quest; there is nothing to share and nothing to fear. Never name a child or an answer.

Privacy checklist

- Quest Logs are never held up to the camera and never read out.
- No answer is said aloud during or after the quest. Gates open for everyone, whatever the answers.
- No clock, countdown or timer is shown; nobody is told who finished first or last.
- Photos come to the program's business WhatsApp (+65 9880 5962) only; tutors never use a personal phone; photos are kept 90 days after the program ends, then deleted; the line goes to the parent only; no private message to any child.
- No score, no x out of 8, no ranking, no tally, no comparison between children, not even in your own notes.
- The recording captures the tutor, the shared screen and any voice that is not muted: the recording is paused for the whole quest (slides s-quest1 and s-quest2) and restarted for the workbook. The WhatsApp number is never on a slide.

Absent, late joiners, families who prefer not to, upset children

- A child who was away or whose family prefers another time: plays the quest at the make-up slot BEFORE watching the recording (the quest is not on the recording, so no gate is given away). It is a small group or a pair with Mi, with a parent beside the child on camera. The tutor never sits alone with one child on a private call.
- A child who joined late: the family may choose to leave out gates for lessons the child did not attend. Write nothing for those gates; if no gate of an area was taught, the line for the grown-up says nothing about that area.
- A family that does not want the quest: no problem. The child does the Lesson 20 work as usual and the line for the grown-up uses your own notes from the lessons only.
- A child who is upset or stuck: remind the group (open session) that every gate can be skipped and Mi swims round it. Do not single the child out. The grown-up may stop the quest at any time.
- A family with no printer: tell the family to say so a week before and arrange another way (for example the make-up slot, with the Quest Log sent to the family another way). Never ask a child about it in front of the group.

4. Answers and what to look for

Use these to understand what you see on camera and in the photos. Nothing is written on the child's paper, nothing is totalled and no gate is marked right or wrong in front of anyone. A child may write steps in any clear layout. Spelling and neatness do not matter.

Gate 1 (L11) Siti is 4th in a line. Ben stands right behind Siti.
Answer: 5th; 3
(a) Ben is __ in the line. = 5th; (b) There are [ ] children in front of Siti. = 3
A secure answer looks like: Both correct (5th or 5 is right for part a). **Half way there:** One of the two correct.
Worked: Right behind the 4th child is the 5th. In front of Siti are the 1st, 2nd and 3rd children: 3 children.
Watch for: (a) 3rd or 4th.; (b) 4 (counting Siti herself).

Gate 2 (L12) Ravi has 26 marbles. Lily has 9 more marbles than Ravi.
Answer: 35; 61
(a) Lily has __ marbles. = 35; (b) They have __ marbles altogether. = 61
A secure answer looks like: Both correct. **Half way there:** Part a right and part b wrong, or part b right from a wrong part a (counts as one).
Worked: Lily has 9 more than Ravi, so add: 26 + 9 = 35. Altogether: 26 + 35 = 61.
Watch for: (a) 17 (taking away).; (b) 35 (stopping after part a) or 35 + 9.

Gate 3 (L13) Look at the ten-frames.
Answer: 18; odd
(a) 9 + 9 = __ = 18; (b) Write odd or even. 15 is __. = odd
A secure answer looks like: Both correct. **Half way there:** One of the two correct.
Worked: 9 + 9 = 18 (one ten-frame is filled with 9, and so is the other: 10 + 8). 15 pairs up with 1 left over, so 15 is odd.
Watch for: (a) 17 or 19 (a counting slip).; (b) even (the last digit 5 is not read).

Gate 4 (L15) Jenny has 3 jars. Each jar has 5 sweets.
Answer: 3 × 5 = 15; 8
(a) Fill in the boxes. [ ] × [ ] = [ ][ ] = 3 × 5 = 15 (or 5 × 3); (b) Count in 2s: 2, 4, 6, __, 10 = 8
A secure answer looks like: Both correct (3 and 5 in either order, and 15). **Half way there:** One of the two correct, or the product right with the factors swapped or missing.
Worked: 3 groups of 5: 5 + 5 + 5 = 15, so 3 × 5 = 15 (5 × 3 = 15 is also right). Counting in 2s: 2, 4, 6, 8, 10.
Watch for: (a) 3 + 5 = 8.; (b) 7 (counting on by 1).

Gate 5 (L17) Siti shares 16 stickers equally among 4 friends. Kumar puts 10 pens into boxes, 2 pens in each box.
Answer: 4; 5
(a) Each of Siti's friends gets [ ] stickers. = 4; (b) Kumar needs [ ] boxes. = 5
A secure answer looks like: Both correct. **Half way there:** One of the two correct.
Worked: 16 shared equally among 4 friends is 4 each. 10 pens in groups of 2 make 5 boxes (2, 4, 6, 8, 10).
Watch for: (a) 12 (16 - 4) or 20.; (b) 8 (10 - 2) or 2 (the pens in each box).

Gate 6 (L18) Mei has 5 bags. Each bag has 3 sweets. She gives 4 sweets to Ben.
Answer: 15; 11
(a) The bags have __ sweets. = 20; (b) Mei has __ sweets left. = 17
A secure answer looks like: Both correct. **Half way there:** Part a right and part b wrong (for example 19), or part b right from a wrong part a.
Worked: 5 groups of 3 is 15 sweets. She gives 4 away: 15 - 4 = 11.
Watch for: (a) 8 (adding 5 and 3).; (b) 19 (adding 4) or 4.

Gate 7 (L19) Each shape is a number. The heart + the heart = 18. Ben says the heart is 8.
Answer: No; 9
(a) Is Ben correct? Circle Yes or No. (Yes / No) = No; (b) The heart is __. = 9
A secure answer looks like: No and 9. **Half way there:** One of the two correct.
Worked: 8 + 8 = 16, not 18, so Ben is not correct. 9 + 9 = 18, so the heart is 9.
Watch for: (a) Yes (not checking by adding 8 + 8).; (b) 18 (the total).

Gate 8 (L20) 8 buns cost $16.
Answer: $2; $10
(a) 1 bun costs $__. = 2; (b) 5 buns cost $__. = 10
A secure answer looks like: Both correct. **Half way there:** Part a right and part b wrong, or part b right from a wrong part a.
Worked: $16 shared equally among 8 buns is $2 each. 5 buns cost 5 groups of $2, which is $10.
Watch for: (a) 24 (adding) or 128.; (b) 16 (the total) or 7 (2 + 5).

Secret cave (optional) (L11) Ravi is 8th from the front and 8th from the back of a line. Mei says there are 16 children in the line.
Answer: No; 15 children (7 in front of Ravi, Ravi, and 7 behind him)
(a) Is Mei correct? Circle Yes or No. (Yes / No) = No; (b) There are __ children in the line. = 15
A secure answer looks like: The answer is as above. **Half way there:** -
Worked: There are 7 children in front of Ravi and 7 behind him, and Ravi himself: 7 + 7 + 1 = 15. 8 + 8 = 16 counts Ravi twice.
Watch for: 16 (counting Ravi twice).; 14 (leaving Ravi out).

5. Tutor private notes (tutor only; one sheet per child; delete when the line is sent)

Child (first name only): ..... Date: ..... Quest partner: .....

Fill this in after the lesson, from the photos and what you saw on camera. Write what the child did and said in your own words. No ticks for right or wrong, no points, nothing added up. The gate numbers are only reminders of where to look.

Gate	From	What I saw (words used, counters or drawing, how the partner helped, anything the child said)
1	L11	
2	L12	
3	L13	
4	L15	
5	L17	
6	L18	
7	L19	
8	L20	
Cave	L11	

Prompt	My notes
Something the child did well (a named strength, for example: drew a strip to find how many more marbles one child had)	
How they started: counters, drawing a strip, ten-frames, or in their head	
Where they paused or turned to their partner, and what helped	
One next step for home (small and concrete)	
Did they try the secret cave? (a neutral note only)	

The one line for the grown-up (section 7), written from these notes:

.....

.....

6. Group advice for next term (tutor only; one sheet per class; no names)

When all the notes are written, look across them without names. Write what several children were still working on and how you will use the first two lessons of next term (a warm-up, a picture, a game). This page never goes to families.

Area	What several children were still working on (no names)	What I will do in the first two lessons of next term
Numbers, adding and taking away Lessons 11 to 19: places in a line (1st, 2nd, 3rd), strips for more and fewer, doubles, odd and even numbers, and finding a hidden number in a picture equation		
Groups, sharing and stories about groups Lessons 15 to 20: groups of the same size and the times sign, counting in 2s and 5s, sharing equally and putting into groups, stories about groups with a second step, and finding the cost of one first		

Prompt	My notes
Gates or ideas where several children paused (no names)	
A warm-up or game to bring back in the first lesson of next term	
What worked well about the quest as a game, to keep next time	

7. The one line for the grown-up: what we saw

Mi's Challenge Quest: [Name]
In Mi's Challenge Quest, [Name] [one specific thing you saw, in plain words]. [Optional: one small next step at home.]

One or two sentences at most. Say what the child did (drew a strip, shared equally, explained a step), not how many answers were right. Use the lines below as patterns and always add your own detail from your notes.

Area (parent wording)	Things you might have seen (pick the one your notes show; add the detail)	A next step that helps at home
Numbers, adding and taking away Lessons 11 to 19. We looked at places in a line (1st, 2nd, 3rd), strips for more and fewer, doubles, odd and even numbers, and finding a hidden number in a picture equation.	• [Name] found a place in a line (1st, 2nd, 3rd) and used a strip to find the other number and the total. • [Name] doubled numbers and told odd from even using the ten-frames. • [Name] tried a friend's number in a picture equation by putting it back, and explained what happened.	• Line up toys at home and say 1st, 2nd, 3rd; draw two strips for more and fewer stories. • Put a number back into a sum to try it, with counters.
Groups, sharing and stories about groups Lessons 15 to 20. We looked at groups of the same size and the times sign, counting in 2s and 5s, sharing equally and putting into groups, stories about groups with a second step, and finding the cost of one first.	• [Name] wrote groups of the same size with the times sign and counted in 2s. • [Name] shared equally and put things into groups, using counters or a drawing. • [Name] solved a story about groups with a second step, and found the cost of one first.	• Share real counters onto plates to find how many on each. • Count groups of the same size with cups and counters, then say the times sentence.

A filled example (fictional child, Mei)

Mi's Challenge Quest: Mei
In Mi's Challenge Quest, Mei drew a strip to find how many more marbles one child had, and shared 16 stickers equally among 4 friends. A good next step: share real counters onto plates at home to find how many in each.

This line describes Mei's own thinking in the game. It does not compare children.

Words to avoid in the line: score, mark, out of, rank, top, below average, behind, ahead of, pass, fail, predicted, ready for. No claim about school exams, GEP or any contest, and no mention of other children. The line goes to the parent only. If the child skipped the secret cave or a gate, say nothing about it. If the child tried the secret cave, you may add: "[Name] also had a go at the secret cave."